

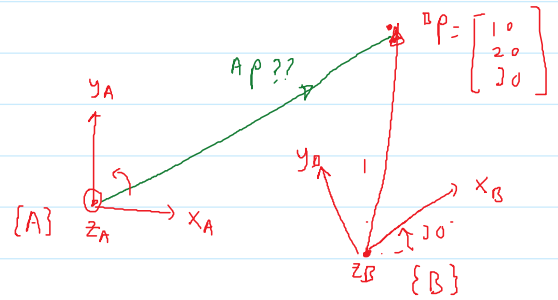
Question 1

- A position vector of a point in frame {B} is given by

$${}^B P = \begin{bmatrix} 10 \\ 20 \\ 30 \end{bmatrix}$$

- The origin of frame {B} is at [11, -3, 9] with respect to frame {A}.
- Also, frame {B} is rotated by 30 degrees along the z-axis of frame {A}.

- What is the position vector of the point in frame {A}?
- Also, write down the homogenous transformation matrix.



1)

$${}^A P = {}^A R_B \cdot {}^B P + {}^A P_{BORG}$$

$${}^A R_B = \begin{bmatrix} c30^\circ & -s30^\circ & 0 \\ s30^\circ & c30^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$${}^A P = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 20 \\ 30 \end{bmatrix} + \begin{bmatrix} 11 \\ -3 \\ 9 \end{bmatrix} = \begin{bmatrix} 9.66 \\ 19.32 \\ 39 \end{bmatrix}$$

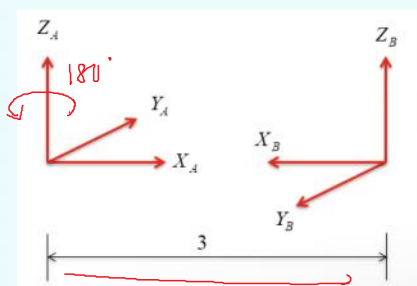
2)

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} A \\ B \end{bmatrix} \cdot \begin{bmatrix} {}^B P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A R_B & {}^A P_{BORG} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^B P \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.866 & -0.5 & 0 & 11 \\ 0.5 & 0.866 & 0 & -3 \\ 0 & 0 & 1 & 9 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 20 \\ 30 \\ 1 \end{bmatrix} = \begin{bmatrix} 9.66 \\ 19.32 \\ 39 \\ 1 \end{bmatrix}$$

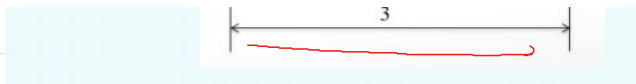
Question 2

- Give the value of ${}^A T_B$.



$${}^A T_B = \begin{bmatrix} c180^\circ & -s180^\circ & 0 & 3 \\ s180^\circ & c180^\circ & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 & 0 & 3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$= \begin{bmatrix} -1 & 0 & 0 & 3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \end{bmatrix}$$

Question 3

fixed angles

- A vector AP is rotated about Z_A by θ degrees.
- It is subsequently rotated about X_A by ϕ degrees.
- Give the rotation matrix that accomplishes these rotations in the specified order.

order of physical rotations
= opposite the order of
matrix multiplication

in video: Physical: $x \rightarrow y \rightarrow z$
Matrix: $R = R_z R_y R_x$

$y \rightarrow z \rightarrow x$
 $R_x R_z R_y$

Physical: $Z \rightarrow X$

Matrix: $R = R_x(\phi) \cdot R_z(\theta)$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\phi & -s\phi \\ 0 & s\phi & c\phi \end{bmatrix} \cdot \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c\theta & -s\theta & 0 \\ c\phi s\theta & c\phi c\theta & -s\phi \\ s\phi s\theta & s\phi c\theta & c\phi \end{bmatrix}$$

Question 4

- A vector AP is rotated about Y_A by 30 degrees.
- It is subsequently rotated about X_A by 45 degrees.
- Give the rotation matrix that accomplishes these rotations in the specified order.

Physical: $Y \rightarrow X$

Matrix $R = R_x(45) R_y(30)$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & c45 & -s45 \\ 0 & s45 & c45 \end{bmatrix} \cdot \begin{bmatrix} c30 & 0 & s30 \\ 0 & 1 & 0 \\ -s30 & 0 & c30 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.707 & -0.707 \\ 0 & 0.707 & 0.707 \end{bmatrix} \cdot \begin{bmatrix} 0.866 & 0 & 0.5 \\ 0 & 1 & 0 \\ -0.5 & 0 & 0.866 \end{bmatrix}$$

$$= \begin{bmatrix} 0.866 & 0 & 0.5 \\ 0.3536 & 0.707 & -0.6123 \\ -0.3536 & 0.707 & 0.6123 \end{bmatrix}$$

Side notes:

Fixed angle (roll, pitch, yaw) $\rightarrow$ aerospace

Euler angles $\rightarrow$ KUKA



(Equivalent angle axis)

$\hookrightarrow$ Euler parameters / Quaternions $\Sigma_1 \dots \Sigma_4 \rightarrow$ ABB



Question 5

Euler angles

- A frame {B} is originally coincident with frame {A}.
- We first rotate {B} about Z_B by θ degrees.
- Then we rotate the resulting frame about X_B by ϕ degrees.
- Give the rotation matrix that will change the descriptions of vectors from ${}^B P$ to ${}^A P$.

Physical order = matrix multiplication order

e.g. video $Z \rightarrow Y \rightarrow X: R = R_Z R_Y R_X$

$$\left(\begin{array}{l} Y \rightarrow X \rightarrow Z : R = R_Y R_X R_Z \\ Z \rightarrow Y \rightarrow Z : R = R_Z(45^\circ) \cdot R_Y(30^\circ) \cdot R_Z(60^\circ) \end{array} \right)$$

Physical: $Z_B \rightarrow X_B$

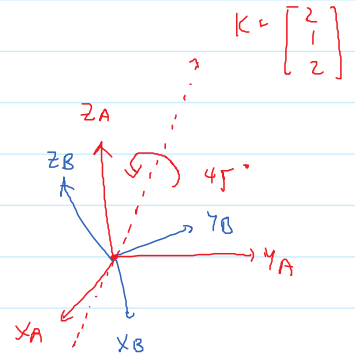
Rotation matrix: $R = R_Z(\theta) \cdot R_X(\phi)$

$$= \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\phi & -s\phi \\ 0 & s\phi & c\phi \end{bmatrix}$$

$$= \begin{bmatrix} c\theta & -s\theta c\phi & s\theta s\phi \\ s\theta & c\theta c\phi & -c\theta s\phi \\ 0 & s\phi & c\phi \end{bmatrix}$$

Question 6

- The axis of a particular rotation is $K = [2, 1, 2]$. (NOTE: this is not yet a unit vector).
- This axis passes through the origin.
- The angle of rotation is 45 degrees.
- Derive the rotation matrix based on equivalent angle-axis representation.



1) K needs to be changed into a unit vector
 $\hookrightarrow$ divide K by magnitude. $|K| = \sqrt{2^2 + 1^2 + 2^2} = 3$

$$\Rightarrow \hat{K} = \frac{K}{|K|} = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \end{bmatrix}^T$$

$\uparrow \quad \uparrow \quad \uparrow$
 $k_x \quad k_y \quad k_z$

2) $\theta = 45^\circ$

$$s\theta = \sin 45^\circ = 0.7071$$

$$c\theta = \cos 45^\circ = 0.7071$$

$$v\theta = 1 - c\theta = 1 - 0.7071 = 0.2929$$

$$3) R = \begin{bmatrix} k_x^2 v\theta + c\theta & - & - \\ k_x k_y v\theta + k_z s\theta & - & - \\ - & - & - \end{bmatrix} = \begin{bmatrix} 0.8373 & -0.4063 & 0.3659 \\ 0.5365 & 0.7396 & -0.4063 \\ -0.1055 & 0.5365 & 0.8373 \end{bmatrix}$$

Double check: check if columns are unit vector

Question 7

- For the same rotations as in Question 6, derive the rotation matrix based on Euler parameters.

$$1) \begin{aligned} \xi_1 &= k_x \sin\left(\frac{\theta}{2}\right) = \frac{2}{3} \sin(22.5^\circ) = 0.2551 \\ \xi_2 &= k_y \sin\left(\frac{\theta}{2}\right) = \frac{1}{3} \sin(22.5^\circ) = 0.1276 \\ \xi_3 &= k_z \sin\left(\frac{\theta}{2}\right) = \frac{2}{3} \sin(22.5^\circ) = 0.2551 \\ \xi_4 &= \cos\left(\frac{\theta}{2}\right) = \cos(22.5^\circ) = 0.9239 \end{aligned}$$

$$\text{Verify } \xi_1^2 + \xi_2^2 + \xi_3^2 + \xi_4^2 = 1$$

$$2) \quad R = \begin{bmatrix} 1 - 2\xi_2^2 - 2\xi_3^2 & \dots & \dots \\ 2(\xi_1\xi_2 + \xi_3\xi_4) & \dots & \dots \\ \dots & \dots & \dots \end{bmatrix} = \text{same answer as } \underline{\text{question 6}}$$

Question 8

- A rotation matrix is given by:

$$R = \begin{bmatrix} 0.8373 & -0.4063 & 0.3659 \\ 0.5365 & 0.7397 & -0.4063 \\ -0.1055 & 0.5365 & 0.8373 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

- Find out all the parameters in terms of:

- a. X-Y-Z fixed angles
- b. Z-Y-X Euler angles
- c. Equivalent Angle-Axis representation
- d. Euler parameters

a) X-Y-Z fixed angles.

$$\begin{aligned} \beta &= \arctan 2 \left(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2} \right) \\ &= \arctan 2 \left(0.1055, \sqrt{0.8373^2 + 0.5365^2} \right) \\ &= \arctan 2 \left(\underbrace{0.1055}_y, \underbrace{0.9944}_x \right) \\ &= \underline{\underline{6.0561^\circ}} \end{aligned}$$

$\arctan\left(\frac{y}{x}\right)$

+	+	-	-
+	-	-	+
+	1	2	3
-			4
	+	-	+

Why divide by $\cos \beta$?

e.g. $\cos \beta < 0$

$\left. \begin{array}{l} r_{21} + \\ r_{11} - \end{array} \right\}$	$\left. \begin{array}{l} + \\ - \end{array} \right\}$	$\left. \begin{array}{l} + \\ - \end{array} \right\}$	$\left. \begin{array}{l} + \\ - \end{array} \right\}$	$\left. \begin{array}{l} + \\ - \end{array} \right\}$
---	---	---	---	---

$\alpha = \arctan\left(\frac{+}{-}, \frac{-}{-}\right)$

$= \arctan(-, +)$

$$\begin{aligned} \alpha &= \arctan 2 \left(\frac{r_{21}}{\cos \beta}, \frac{r_{11}}{\cos \beta} \right) \\ &= \arctan 2 \left(\frac{0.5365}{0.9944}, \frac{0.8373}{0.9944} \right) \\ &= \arctan 2 (0.5395, 0.8420) \\ &= \underline{\underline{32.6492^\circ}} \end{aligned}$$

$$\begin{aligned} \gamma &= \arctan 2 \left(\frac{r_{22}}{\cos \beta}, \frac{r_{12}}{\cos \beta} \right) \\ &= \arctan 2 \left(\frac{0.5365}{\cos \beta}, \frac{0.8373}{\cos \beta} \right) \\ &= \underline{\underline{32.6492^\circ}} \end{aligned}$$

In summary, R is obtained from:
1st rotate 32.6492° about X_A

In summary, R is obtained from:

1st rotate 32.6492° about X_A

2nd rotate 6.0561° about Y_A

3rd rotate 32.6492° about Z_A

b) $Z \rightarrow Y \rightarrow X$ Euler angles:

We know that this is equivalent to $X \rightarrow Y \rightarrow Z$ fixed angles!

The R matrix is obtained from

1st rotate 32.6492° about Z_B

2nd rotate 6.0561° about Y_B

3rd rotate 32.6492° about X_B

c) Equivalent angle axis

$$\theta = \arccos \left(\frac{r_{11} + r_{22} + r_{33} - 1}{2} \right)$$

$$= \arccos \left(\frac{0.8373 + 0.7397 + 0.8373 - 1}{2} \right)$$

$$= \arccos(0.7071) = \underline{\underline{45^\circ}}$$

$$\hat{k} = \frac{1}{2 \sin \theta} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix}$$

$$= \frac{1}{2 \sin 45^\circ} \begin{bmatrix} 0.5365 - (-0.4063) \\ 0.3659 - (-0.1055) \\ 0.5365 - (-0.4063) \end{bmatrix}$$

$$= \begin{bmatrix} 0.66 \\ 0.33 \\ 0.66 \end{bmatrix} = \underline{\underline{\begin{bmatrix} 2/3 \\ 1/3 \\ 2/3 \end{bmatrix}}}$$

d) Euler parameters / Quaternions

$$\mathcal{E}_4 = \frac{1}{2} \sqrt{1 + r_{11} + r_{22} + r_{33}}$$

$$= \frac{1}{2} \sqrt{1 + 0.8373 + 0.7397 + 0.8373}$$

$$= \frac{1}{2} \sqrt{1 + 0.8873 + 0.7397 + 0.8873}$$

$$= \underline{\underline{0.9239}}$$

$$\varepsilon_1 = \frac{r_{32} - r_{23}}{4\varepsilon_4} = \underline{\underline{0.2551}}$$

$$\varepsilon_1 = \frac{1}{2} \sqrt{1 - + -}$$

$$\varepsilon_2 = \frac{r_{13} - r_{31}}{4\varepsilon_4} = \underline{\underline{0.1276}}$$

$$\varepsilon_3 = \frac{r_{21} - r_{12}}{4\varepsilon_4} = \underline{\underline{0.2551}}$$

Question 9

- The following frame definitions are given as known.

$$\begin{pmatrix} u \\ A \end{pmatrix}^T = \begin{bmatrix} 0.866 & -0.5 & 0 & 11 \\ 0.5 & 0.866 & 0 & -1 \\ 0 & 0 & 1 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \begin{pmatrix} B \\ A \end{pmatrix}^T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.866 & -0.5 & 10 \\ 0 & 0.5 & 0.866 & -20 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{pmatrix} C \\ U \end{pmatrix}^T = \begin{bmatrix} 0.866 & -0.5 & 0 & -3 \\ 0.433 & 0.75 & -0.5 & -3 \\ 0.25 & 0.433 & 0.866 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Solve for $\begin{pmatrix} B \\ C \end{pmatrix}^T$.

$$\begin{pmatrix} B \\ C \end{pmatrix}^T = \begin{pmatrix} B \\ A \end{pmatrix}^T \cdot \begin{pmatrix} A \\ U \end{pmatrix}^T \cdot \begin{pmatrix} U \\ C \end{pmatrix}^T$$

(1) (2)

$$\begin{pmatrix} A \\ B \end{pmatrix}^T = \left[\begin{array}{ccc|c} \begin{pmatrix} A \\ B \end{pmatrix}^T & -\begin{pmatrix} A \\ B \end{pmatrix}^T \begin{pmatrix} A \\ B \end{pmatrix}^T \end{array} \right]$$

$$1) \begin{pmatrix} U \\ A \end{pmatrix}^T = \left[\begin{array}{ccc|c} \begin{pmatrix} U \\ A \end{pmatrix}^T & -\begin{pmatrix} U \\ A \end{pmatrix}^T \begin{pmatrix} U \\ A \end{pmatrix}^T \end{array} \right]$$

$$= \left[\begin{array}{ccc|c} 0.866 & -0.5 & 0 & -9.026 \\ -0.5 & 0.866 & 0 & 1.316 \\ 0 & 0 & 1 & -8 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

$$= \left[\begin{array}{ccc|c} 0.866 & -0.5 & 0 & -9.026 \\ -0.5 & 0.866 & 0 & 1.316 \\ 0 & 0 & 1 & -8 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

$$\begin{pmatrix} C \\ U \end{pmatrix}^T = \left[\begin{array}{ccc|c} \begin{pmatrix} C \\ U \end{pmatrix}^T & -\begin{pmatrix} C \\ U \end{pmatrix}^T \begin{pmatrix} C \\ U \end{pmatrix}^T \end{array} \right]$$

$$= \left[\begin{array}{ccc|c} 0.866 & 0.433 & 0.25 & -0.866 \\ -0.433 & -0.25 & -0.866 & -3 \end{array} \right]$$

$$= \left[\begin{array}{ccc|ccc} 0.866 & 0.433 & 0.25 & -0.866 & -0.433 & -0.25 \\ -0.5 & 0.75 & 0.433 & 0.5 & -0.75 & -0.433 \\ 0 & -0.5 & 0.866 & 0 & 0.5 & -0.866 \\ \hline 0 & 0 & 0 & & & \end{array} \right] \begin{bmatrix} -3 \\ -3 \\ 3 \end{bmatrix}$$

$$= \left[\begin{array}{ccc|c} 0.866 & 0.433 & 0.25 & 3.147 \\ -0.5 & 0.75 & 0.433 & -0.549 \\ 0 & -0.5 & 0.866 & -4.098 \\ \hline 0 & 0 & 0 & 1 \end{array} \right]$$

Finally, plug $\tilde{A}^{-1}$ & $\tilde{C}^{-1}$ into $\underline{\underline{CT = \tilde{C}^{-1} \cdot \tilde{A}^{-1} \cdot C^T}} \rightarrow \text{final answer}$